Linear Algebra | Quiz for §6.1 | November 13, 2026

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Directions: Show and explain all work. Put phones away. No calculators, etc.

1. Suppose $T : \mathbb{R}^3 \to \mathbb{R}^2$ is a linear transformation for which:

$$T\left(\begin{bmatrix}1\\1\\1\end{bmatrix}\right)=\begin{bmatrix}0\\-1\end{bmatrix},\quad T\left(\begin{bmatrix}0\\-2\\2\end{bmatrix}\right)=\begin{bmatrix}-1\\2\end{bmatrix},\quad \text{and}\quad T\left(\begin{bmatrix}0\\-1\\2\end{bmatrix}\right)=\begin{bmatrix}2\\-1\end{bmatrix}.\quad \text{Find } T\left(\begin{bmatrix}4\\2\\0\end{bmatrix}\right).$$

First, let's find $\begin{bmatrix}4\\2\\0\end{bmatrix}$ as a linear combination of $\begin{bmatrix}1\\1\\1\end{bmatrix}, \begin{bmatrix}0\\-2\\2\end{bmatrix}, \begin{bmatrix}0\\-1\\2\end{bmatrix}$, that is, find x, y and z for which $\begin{bmatrix}4\\2\\0\end{bmatrix} = x\begin{bmatrix}1\\1\\1\end{bmatrix} + y\begin{bmatrix}0\\-2\\2\end{bmatrix} + z\begin{bmatrix}0\\-1\\2\end{bmatrix}$.

$$\Rightarrow \begin{bmatrix}x\\x-2y-z\\x+2y+2z\end{bmatrix} = \begin{bmatrix}4\\2\\0\end{bmatrix} \Rightarrow \text{solve } \begin{cases} x = 0\\ x-2y-z=0\\ x+2y+2z=0\end{cases}$$

$$\left[\begin{array}{ccc|c}1&0&0&4\\1&-2&-1&2\\1&2&2&0\end{array}\right] \xrightarrow{-R_1+R_2\to R_2,\ -R_1+R_3\to R_3} \left[\begin{array}{ccc|c}1&0&0&4\\0&-2&-1&-2\\0&2&2&-4\end{array}\right] \xrightarrow{R_2+R_3\to R_3} \left[\begin{array}{ccc|c}1&0&0&4\\0&-2&-1&-2\\0&0&1&-6\end{array}\right]$$

$$\xrightarrow{R_3+R_2\to R_2} \left[\begin{array}{ccc|c}1&0&0&4\\0&-2&0&-8\\0&0&1&-6\end{array}\right] \xrightarrow{-\frac{1}{2}R_2\to R_2} \left[\begin{array}{ccc|c}1&0&0&4\\0&1&0&4\\0&0&1&-6\end{array}\right] \quad \text{Thus } \begin{cases} x=4\\ y=4\\ z=-6\end{cases}$$

and so $\begin{bmatrix}4\\2\\0\end{bmatrix} = 4\begin{bmatrix}1\\1\\1\end{bmatrix} + 4\begin{bmatrix}0\\-2\\2\end{bmatrix} - 6\begin{bmatrix}0\\-1\\2\end{bmatrix}$

Now $T\left(\begin{bmatrix}4\\2\\0\end{bmatrix}\right) = T\left(4\begin{bmatrix}1\\1\\1\end{bmatrix} + 4\begin{bmatrix}0\\-2\\2\end{bmatrix} - 6\begin{bmatrix}0\\-1\\2\end{bmatrix}\right)$

$$= 4T\left(\begin{bmatrix}1\\1\\1\end{bmatrix}\right) + 4T\left(\begin{bmatrix}0\\-2\\2\end{bmatrix}\right) - 6T\left(\begin{bmatrix}0\\-1\\2\end{bmatrix}\right)$$

$$= 4\begin{bmatrix}0\\-1\end{bmatrix} + 4\begin{bmatrix}-1\\2\end{bmatrix} - 6\begin{bmatrix}2\\-1\end{bmatrix}$$

$$= \begin{bmatrix}0-4-12\\-4+8+6\end{bmatrix} = \boxed{\begin{bmatrix}-16\\10\end{bmatrix}} \leftarrow \text{Answer}$$